

Optimizer — AI Feature Audit

Status of every AI feature against the current codebase — 43 features across 10 categories.

COMPLETED	15
AI PLANNER	Automatic Layout Generation
AI PLANNER	Smart Tower Placement
AI CIVIL ENGINEER	Civil Design Suggestions
AI STRUCTURAL ADVISOR	Column Grid Suggestions
AI STRUCTURAL ADVISOR	Foundation Recommendations
AI STRUCTURAL ADVISOR	Structural Risk Warnings
AI COST ENGINEER	Cost Prediction
AI COST ENGINEER	Cost Comparison
AI COST ENGINEER	Cost Saving Suggestions
AI SUSTAINABILITY	Green Building Score
AI SUSTAINABILITY	Water Efficiency
RECOMMENDATION ENGINE	Design Improvements
RECOMMENDATION ENGINE	Engineering Suggestions
SCENARIO COMPARISON	Compare Costs
SCENARIO COMPARISON	Compare Parking

INCOMPLETE	15
AI PLANNER	Apartment Mix Optimisation
AI PLANNER	Floor Optimisation
AI PLANNER	Open Space Optimisation
AI CIVIL ENGINEER	FAR Optimisation
AI CIVIL ENGINEER	FSI Optimisation
AI CIVIL ENGINEER	Utility Optimisation
AI CIVIL ENGINEER	Parking Optimisation
AI STRUCTURAL ADVISOR	Beam Layout Suggestions
AI QUANTITY SURVEYOR	Quantity Optimisation
AI QUANTITY SURVEYOR	BOQ Optimisation
AI COST ENGINEER	Budget Optimisation
AI SUSTAINABILITY	Solar Potential
RECOMMENDATION ENGINE	Cost Optimisation
RECOMMENDATION ENGINE	Material Recommendations
SCENARIO COMPARISON	Compare Layouts

MISSING	13
AI QUANTITY SURVEYOR	Waste Reduction
AI SUSTAINABILITY	Carbon Footprint
AI SUSTAINABILITY	Tree Plantation Suggestions
AI BUSINESS CONSULTANT	ROI Analysis
AI BUSINESS CONSULTANT	Profit Forecast
AI BUSINESS CONSULTANT	Project Feasibility
AI BUSINESS CONSULTANT	Break-even Analysis
AI CHAT ASSISTANT	Project Q&A
AI CHAT ASSISTANT	Civil Engineering Guidance
AI CHAT ASSISTANT	Report Explanation
AI CHAT ASSISTANT	Scenario Comparison (chat)
SCENARIO COMPARISON	Compare Sustainability
SCENARIO COMPARISON	Compare ROI

Completed — implemented and working. **Incomplete** — the value is computed and checked, but nothing optimises it; the app reports what a scheme *is*, not what a better one would be. **Missing** — no code exists.

The 13 missing items cluster into three buildable blocks: a **financial layer** (ROI, profit, feasibility, break-even, Compare ROI — one revenue model unlocks all five), a **chat assistant** (all four items — one endpoint plus a UI panel), and **carbon, waste, trees and Compare Sustainability**.